Lab Assignment - 1: DBMS Odd-2026

Vaibhav Sharma (2501030279)

August 2, 2026

Answer 1

Entities Involved :

1. **Books:** Book_ID (Primary Key), ISBN, Title, Author (Foreign Key), Publication_Year, Copies, Genre
2. **Members:** Member_ID (Primary Key), Name, Email, Membership_Date, Validity
3. **Authors:** Author_ID (Primary Key), Name, Nationality, Email
4. **Transactions:** Transaction_ID (Primary Key), Book_ID (Foreign Key), Member_ID (Foreign Key), Issue_Date, Return_Date

Answer 2

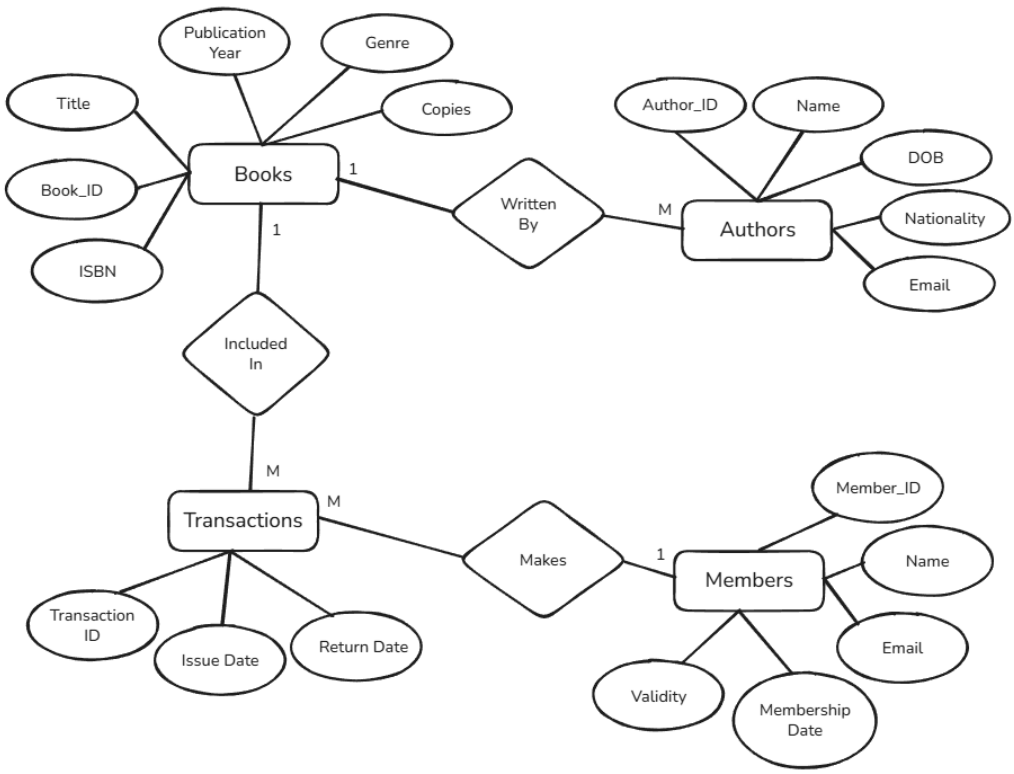

Figure 1: ER Diagram for Library Management System (*Made using Excalidraw)*

Answer 3

1. **Entity Integrity Constraint:** Each entity has a unique primary key (Book_ID, Member_ID, Author_ID) which cannot be NULL.

2. **Referential Integrity Constraint:** Foreign keys are referring to already existing primary keys.

3. **Key Constraint:** Every primary key value will be unique. Ex. IDs are unique for each tuple.

4. **Domain Constraint:** Attributes will contain only valid values. Emails will contain varchar, IDs will contain integer, Name contains only char etc.

Answer 4

1. A new entity, Reservation, to keep track of reserved books.

2. M:N relationship between authors and books, 1:M for members and reservations, 1:M for books and reservations

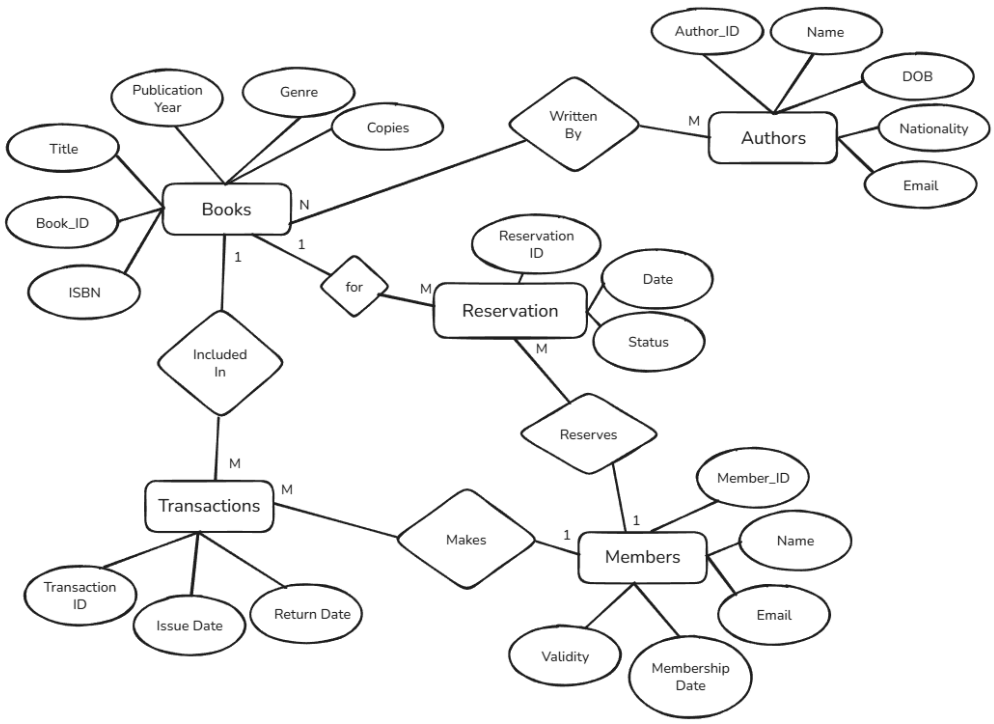

Figure 2: Modified ER Diagram (*Made using Excalidraw*)

Answer 5

Design Errors:

- Order entity has no Primary Key.
- Customer contains Age attribute which is updated every year manually.
- No separate entity is there between Order and Product.

Corrections:

- Order_ID should be the Primary Key.
- DOB instead of Age.
- Order_Product entity between Order and Product.

Affected Constraints:

- No Primary Key: Entity Integrity Constraint.
- Age attribute: Domain Constraint.
- No Order-Product relationship: Referential Integrity Constraint.

Answer 6

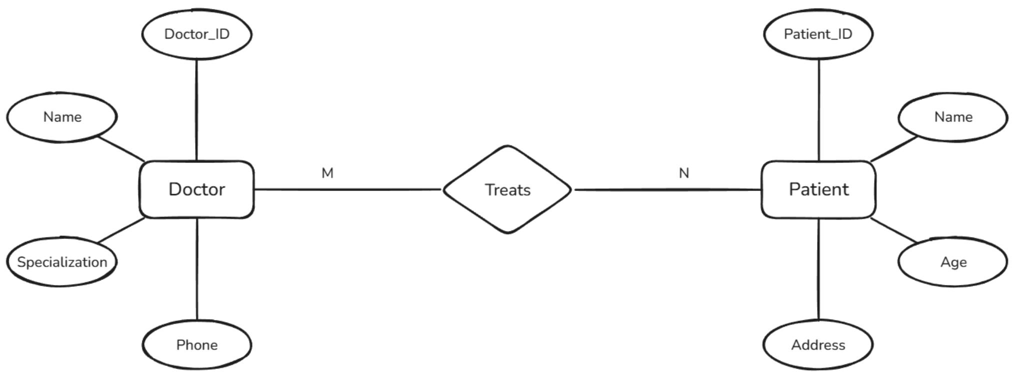

Figure 3: Hospital Management System ER Diagram (*Made using Excalidraw*)

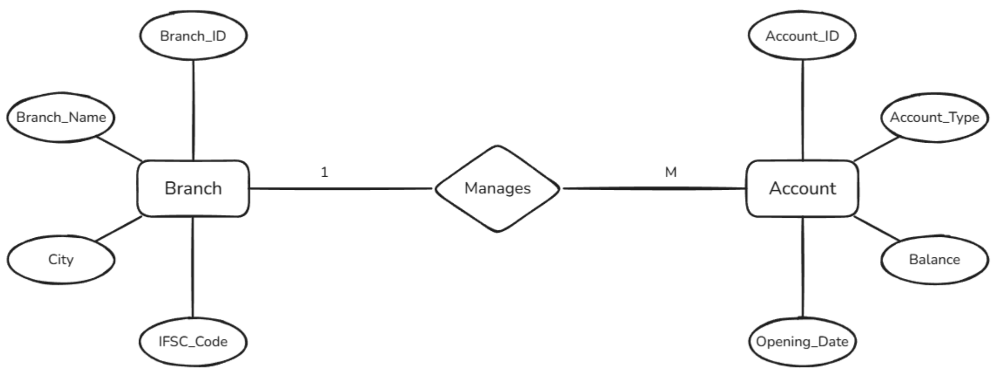

Figure 4: Banking System ER Diagram (*Made using Excalidraw*)

Comparison:
- Hospital System: Doctor and Patient have M:N relationship.
- Banking System: Branch and Account have 1:M relationship.

Integrity Constraints:
- Entity: Doctor, Patient, Branch and Account has a unique Primary Key.
- Referential: Every account and record belongs to existing branch and patient.